

BASED ON THE PAPER

Grant Selection Design and Start-Up Innovation

Evidence from Design-Dependent Subsidy Effects

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Curriculum Vitae

▶ School

Sidcot Independent School England	2012–2014
International Baccalaureate	

▶ University

Technical University of Munich B.Sc. Technology and Management	2014–2020
Technical University of Munich M.Sc. Technology and Management	2020–2024

▶ Professional Experience

ColInvest Corporate Finance Analyst	2021–2024
agentix GmbH Co-Founder & COO	2024–2025
Independent Management Consultant	2024–present

Standardising convertible loan agreements for start-ups

- ▶ analysed convertible loans as a financing instrument for young firms
- ▶ compared legal, financial, and contractual design choices
- ▶ assessed whether standardisation can reduce complexity and transaction costs
- ▶ developed a model convertible loan agreement

Link to the PhD proposal: how institutional design shapes financing, incentives, and firm behaviour.

Why this PhD topic fits my profile

My background connects **firm finance**, **innovation**, and **public funding**.

- ▶ **Master's thesis:** empirical work on firm finance and investment
- ▶ **agentix GmbH:** Co-Founder and COO of a Munich-based AI automation start-up
- ▶ **EXIST grant:** direct exposure to public start-up funding

Do the signals used to select firms generate the value public funding is meant to create?

- ▶ Which observable signals enter the funding decision?
- ▶ Do these signals predict innovation, spillovers, and economic value?
- ▶ Do programmes weight signals differently when they define value differently?

The project separates the **selection mechanism** from the grant itself.

Theoretical framework: selection signals as policy instruments

The project treats selection signals as **proxies for expected public value**.

Policy objective → Selection signals → Funding decision → Outcome layers

- ▶ Programmes define value differently: innovation, spillovers, or local economic value.
- ▶ Granting authorities translate these objectives into signal weights.
- ▶ The empirical question is whether these weights predict the outcomes they are meant to generate.

The grant is therefore not the only treatment; the selection rule is part of the mechanism.

Empirical framework



The unit of analysis is the **firm–application**. The project links three layers of data:

- ▶ **Selection signals:** founder, project, technology, and regional characteristics
- ▶ **Selection rule:** how programmes weight these signals in the funding decision
- ▶ **Outcome layers:** innovation, spillover potential, and economic value

The empirical object is not the grant alone, but the link between **what programmes select for** and **what selected firms later generate**.

Identification strategy

Identification uses **variation in selection rules**.

- ▶ **Main:** selection model + panel outcome analysis
- ▶ **Cleanest causal design:** RDD at rankings or score cutoffs
- ▶ **Extensions:** Diff-in-Diff / event study where timing varies
- ▶ **Fallback:** IV only with credible external variation

Goal: estimate the **selection–outcome alignment gap**.

Expected contribution

The project contributes to the economics of **public innovation policy design**.

- ▶ **Empirical:** evidence on which selection signals predict public value
- ▶ **Conceptual:** separates the selection rule from the grant itself
- ▶ **Policy-relevant:** identifies gaps between programme objectives and outcomes

The first chapter can become a focused study of the **selection–outcome alignment gap**.

Backup: Data Sources

- ▶ Programme documents, eligibility criteria, scoring rubrics and beneficiary lists.
- ▶ Applicant-level award data from CORDA / ORBIS where accessible.
- ▶ Patent data from EPO / PATSTAT.
- ▶ Firm accounts, business registers and regional employment data.
- ▶ Score-level data from programmes such as the EIC Accelerator where available.

Backup: Identification Risks

- ▶ Self-selection into application.
- ▶ Score manipulation near cut-offs.
- ▶ Missing or incomplete score-level data.
- ▶ Survivorship bias in firm-level outcomes.
- ▶ Outcome correlation across innovation, spillover and economic-value layers.